Date: 26 August 2025

To,

The Chairman

Institutional Review Board (IRB)

Satkhira Medical College

Satkhira

Subject: Application for Institutional Review Board (IRB) Clearance

Dear Sir,

I am writing to formally submit my research protocol entitled **"Prevalence of Dyslipidemia and Hypertension among Type 2 Diabetes Mellitus (T2DM) Patients in a Coastal Region of Bangladesh"** for ethical review and approval by the Institutional Review Board.

The proposed study aims to determine the prevalence and associated risk factors of hypertension and dyslipidemia among patients with type 2 diabetes mellitus in the coastal region of Bangladesh. This research is of significant public health importance and will contribute valuable data for improving management strategies for diabetic patients in the region.

The research will be conducted at Satkhira Medical College Hospital. All ethical standards involving human participants will be strictly maintained, and informed consent will be obtained from each participant prior to inclusion in the study.

Enclosed with this letter is the complete research protocol for your review. I kindly request you to review and grant approval to conduct this study.

Thank you for your time and consideration.

Sincerely,

Dr. Sanjoy Kumar Ray

Principal Investigator (PI)

Resident Physician,

Satkhira Medical College Hospital, Satkhira

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Research protocol

Titile: “Prevalence of Dyslipidemia and Hypertension Among Type 2 Diabetes Mellitus (T2DM) Patients in A Coastal Region of Bangladesh”

Researcher:

Dr. Sanjoy Kumar Ray

Resident Physician,

Satkhira Medical College Hospital, Satkhira

01. Background and Rationale

Satkhira, a coastal district in southwestern Bangladesh, faces escalating health challenges due to rapid lifestyle transitions, inadequate access to healthcare, and unique environmental stressors such as high salinity and periodic food insecurity. These factors have contributed to a growing burden of non-communicable diseases, notably Type 2 Diabetes Mellitus (T2DM), which is now recognized as a major public health issue in Bangladesh[1-3]

Among patients with T2DM, the prevalence of major metabolic comorbidities—hypertension and dyslipidemia—is alarmingly high. Multiple recent studies from Bangladesh report dyslipidemia in T2DM patients affecting more than 70–80% of the population, and hypertension rates ranging from 49% to over 60%[1-5]. For example, a nationwide study found the prevalence of dyslipidemia among lipid-lowering drug-naïve T2DM patients to be 74.2%[1], while other reports document rates as high as 88.7% for any dyslipidemia and 65–77% for individual lipid abnormalities (such as hypertriglyceridemia or low HDL cholesterol)[3-5]. Hypertension, another critical comorbidity, is documented in about 49–64% of T2DM patients in South Asia[5,6].

Key drivers of these comorbidities in Bangladesh include increased age, female gender, obesity (high BMI and central adiposity), sedentary lifestyles, poor dietary patterns, and long-standing diabetes[2,3,7,8,9]. In coastal and rural areas like Satkhira, the risk is further compounded by socioeconomic barriers and limited preventive care access[1, 10].

Despite these challenges, evidence on the local epidemiology of hypertension and dyslipidemia in diabetic patients from the Satkhira district remains scarce. Satkhira Medical College Hospital (SMCH), as the only tertiary care center, is uniquely positioned to provide insights into the risk profile and health status of this vulnerable population[10].

Systematic characterization of these comorbidities among T2DM patients at SMCH is thus essential for evidence-based healthcare planning, targeted intervention, and resource allocation. Understanding local prevalence patterns and risk profiles will not only guide clinical management but also inform public health strategies to address the emerging epidemic of diabetes and its complications in Satkhira and similar coastal regions[1, 2, 3, 5, 10].

02. Research Question

What is the prevalence of dyslipidemia and hypertension among patients with Type 2 Diabetes Mellitus attending Satkhira Medical College Hospital in the coastal region of Bangladesh?

03. Objectives

03.1 General Objective

To assess the prevalence of dyslipidemia and hypertension among individuals with type 2 diabetes mellitus in Satkhira Medical College Hospital.

03.2 Specific Objectives

- To determine the prevalence of hypertension among patients with T2DM.
- To determine the prevalence of dyslipidemia among patients with T2DM.
- To describe the sociodemographic, clinical, and lifestyle factors associated with these comorbidities
- To evaluate the treatment patterns (antidiabetic, antihypertensive, and lipid-lowering agents) and adherence.
- To explore the relationship between glycemic control (HbA1c) and comorbidity status.

04. Methodology

04.1 Study Design

A descriptive cross-sectional study.

04.2 Study Area

The study will be conducted at the Outpatient Departments of Satkhira Medical College Hospital, located in Satkhira district—a coastal area in southwestern Bangladesh.

04.3 Study Population

Patients attending Satkhira Medical College Hospital with a previous diagnosis of T2DM will be invited to participate.

04.4 Inclusion Criteria

- Diagnosed with T2DM at least 6 months prior to data collection.
- Aged 30 years or above.
- Residing in Satkhira district for at least 1 year.
- Consenting to participate.

04.5 Exclusion Criteria

- Critically ill or hospitalized patients.
- Pregnant or lactating women.
- Type 1 diabetes mellitus patients.

04.6 Sample Size Calculation

For unknown prevalence, a value of 50% can be used as a conservative estimate[11, 12].

Using a conservative prevalence (p) of 50% to ensure maximum sample size:

$$n = \frac{Z^2 \cdot p \cdot (1-p)}{d^2} = \frac{(1.96)^2 \cdot 0.5 \cdot 0.5}{(0.07)^2} \approx 196$$

Where:

n= minimum sample size required

Z= Z-value (1.96 for 95% confidence level)

p= estimated prevalence

d= degree of accuracy or margin of error

Considering 10% non-response[13]: Final sample size = ~220 participants

For adjusting for expected non-response, standard methodology is to inflate the calculated sample size by dividing by (1 – non-response rate), as detailed in health research manuals and methodological guides[13].

04.7 Sampling Technique

Non-randomized purposive sampling: purposive sampling of eligible T2DM patients attending out-patient department of Satkhira medical college hospital during the data collection period.

04.8 Data Collection Tools and Variables

- Structured questionnaire: socio-demographic data, lifestyle, diet, occupation, physical activity, smoking.
- Clinical measurements: Height, weight, BMI, waist circumference, blood pressure.
- Laboratory investigations: HbA1c, FBS/2HABF/RBS, Lipid profile (Total cholesterol, LDL-C, HDL-C, triglycerides).

04.9 Operational definitions

Diabetes mellitus[14]:

The key laboratory values for diabetes mellitus diagnosis and monitoring according to the American Diabetes Association (ADA) 2025 Standards of Care include:

- **Hemoglobin A1C (HbA1c)**
    - Diagnostic threshold: ≥6.5% (48 mmol/mol) for diabetes diagnosis
    - A1C goal often individualized but commonly <7% for many adults with diabetes
    - Reflects average blood glucose over approximately 3 months

- **Fasting Plasma Glucose (FPG)**
    - Diabetes diagnosis: FPG ≥126 mg/dL (7.0 mmol/L)
    - Normal fasting glucose is below 100 mg/dL (5.6 mmol/L)

- **Random Plasma Glucose:**
    - Diabetes diagnosis: ≥200 mg/dL (11.1 mmol/L) in presence of symptoms of hyperglycemia

- **Continuous Glucose Monitoring (CGM)**
    - Recommended for adults with type 2 diabetes on glucose-lowering agents beyond insulin for better glucose control insights
    - Increasingly used as part of diabetes management and education

Hypertension[15]:

- SBP ≥140 mmHg and/or DBP ≥90 mmHg or on antihypertensive medication.

Dyslipidemia[16, 17]: Any of the following

- Total cholesterol >200 mg/dL
- LDL >130 mg/dL,
- HDL <40 mg/dL (men), <50 mg/dL (women),
- TG >150 mg/dL.

04.10 Data Analysis Plan

- Data entry and cleaning using SPSS version 22.0 (IBM Corp.).
- Descriptive statistics: mean, SD, frequency, and percentage.
- Bivariate analysis (Chi-square, t-tests) to assess associations.
- Multivariate logistic regression will be used to adjust for potential confounders such as age, sex, BMI, duration of diabetes, etc
- The technical matter of editing, encoding and computerization will be followed up by Investigator.
- Statistical significance set at $p < 0.05$

04.11 Randomization and blinding methods:

Not applicable

04.12 Equipment

- Informed written consent.
- A Structured questionnaire containing items to elicit socio-demographic information (e.g. age, gender, resident, occupation, monthly income, level of education etc.) and relevant information about co-occurring physical illnesses.
- Information regarding lab values of blood glucose, HbA1C and lipid profie.

04.13 Flow chart showing the sequence of tasks

Selection of patients on the basis of inclusion and exclusion criteria.

⇩

Taking informed written consent

⇩

220 subjects will be enrolled

⇩

Interview will be taken from each patients

⇩

Structured questionnaire form containing:

- Demographic data.
- Clinical data regarding diabetes mellitus, hypertension and dyslipidemia
- Measuring of FBG/RBS/HbA1c and lipid profile

⇩

Completion of data collection

⇩

Statistical analysis by SPSS 22.0

⇩

Result and Publication

04.14 Ethical Considerations

Ethical approval will be obtained from the Institutional Review Board (IRB) of Satkhira medical college. The study will maintain all the ethical principles in all procedures according to the declaration of Helsinki. The aim and objectives of the study along with its procedure, risks and benefits will be explained to the study subjects in an essay understandable local Language. The patients and/or responsible family member will be informed about the potential risk of checking blood glucose and lipid profile, which is minimal and if any complication develops that will be managed by experienced physician. Participation in this study will be voluntary and there will be no financial contribution for participation in this study. Informed written consent will be obtained from each participant. Participants will be informed of their results and referred to a doctor if needed. All data will be anonymized and securely stored. This study does not involve any drugs, placebo, organs, tissues, the fetus or the aborts.

05. Timeline

The project will be completed over 4 months. Below is a summary of key activities and their duration:

Activity	Month 1	Month 2	Month 3	Month 4
Ethical approval				
Training of data collectors				
Data collection				
Data entry and analysis				
Report writing and dissemination				

06. Budget Estimate

Estimated budget required for conducting the research:

Item	Unit Cost (BDT)	Quantity	Total (BDT)
Lab tests (FBS/RBS, HbA1c, Lipid profile)	(100+300+400=) 800	220	176,000
Stationery and printing	20	220	4,400
Data entry and analysis support	10,000	1	10,000
Conveyance	-	-	2,000
Miscellaneous/contingency	-	-	5,000

Total Estimated Budget: 197,400 BDT

07. Sources of funding

Currently self-funded, with potential for institutional or donor support.

08. Expected Outcomes and Impact

- Baseline prevalence of hypertension and dyslipidemia in coastal diabetic patients.
- Identification of treatment gaps and poor medication adherence.
- Evidence to support public health intervention planning.
- May contribute to regional or national guideline modifications or pilot screening initiatives.

09. Limitations section

As a cross-sectional study, this research cannot establish causal relationships between variables such as glycemic control and the presence of comorbidities like hypertension and dyslipidemia[18]. The use of a non-random, purposive convenient sampling technique may introduce selection bias, limiting the generalizability of findings to the broader diabetic population in the coastal region[19]. Self-reported data (e.g., lifestyle, medication adherence) may be subject to recall or social desirability bias[20]. Resource constraints may limit the scope of laboratory testing and follow-up, and the study does not include a control or comparator group[21]. Despite these limitations, the study will provide essential baseline epidemiological data for future research and health interventions[22].

10. References

1. Kamrul-Hasan ABM, Alam MS, Nusrat Z, et al. Prevalence and patterns of dyslipidemia among lipid-lowering drug-naïve patients with type 2 diabetes mellitus – A countrywide study in Bangladesh. Endocrinol Diabetes Metab. 2023;6(1):e00329.
2. Mostofa MG, Ferdous NEJ, Bhuiyan MAQ. Prevalence of Dyslipidemia among Type 2 Diabetes Mellitus Patient in Combined Military Hospital, Sylhet. J. Armed Forces Med. Coll. [Internet].

2023 Aug. 21 [cited 2025 Aug. 20];18(2):03-6.
https://www.banglajol.info/index.php/JAFMC/article/view/63985

3. Ahmmed MS, Shuvo SD, Paul DK, Karim MR, Kamruzzaman M, Mahmud N, et al. (2021) Prevalence of dyslipidemia and associated risk factors among newly diagnosed Type-2 Diabetes Mellitus (T2DM) patients in Kushtia, Bangladesh. PLOS Glob Public Health 1(12): e0000003. https://doi.org/10.1371/journal.pgph.0000003
4. Yadav D, Mishra M, Tiwari A, Bisen PS, Goswamy HM, Prasad GB. Prevalence of dyslipidemia and hypertension in Indian type 2 diabetic patients with metabolic syndrome and its clinical significance. Osong Public Health Res Perspect. 2014;5(3):169-75. doi:10.1016/j.phrp.2014.04.009
5. Ali N, Samadder M, Kathak RR, Islam F. Prevalence and factors associated with dyslipidemia in Bangladeshi adults. *PLoS One.* 2023;18(1):e0280672. Published 2023 Jan 20. doi:10.1371/journal.pone.0280672
6. Das H, Banik S. Prevalence of dyslipidemia among the diabetic patients in southern Bangladesh: A cross-sectional study. Diabetes Metab Syndr. 2019;13(1):252-7. doi:10.1016/j.dsx.2018.09.006
7. Kelemework, B., Woubshet, K., Tadesse, S. A., Eshetu, B., Geleta, D., & Ketema, W. (2024). The Burden of Dyslipidemia and Determinant Factors Among Type 2 Diabetes Mellitus Patients at Hawassa University Comprehensive Specialized Hospital, Hawassa, Ethiopia. Diabetes, Metabolic Syndrome and Obesity, *17*, 825–32. https://doi.org/10.2147/DMSO.S448350
8. Worku Misganaw Kebede, Kefyalew Dagne Gizachew, Getaneh Baye Mulu. Prevalence and Risk Factors of Dyslipidemia among Type 2 Diabetes Patients at Debre Berhan Referral Hospital, Amhara, Ethiopia, 2020: A Cross-sectional Study. Ethiop J Health Sci. 2021;31(6):1267. doi:http://dx.doi.org/10.4314/ejhs.v31i6. 23
9. Uddin MJ, Begum MR, Hossain MF. Dyslipidemia and Hypertension among Type 2 Diabetics at a Tertiary Hospital. Bangladesh Med J. 2018;47(4):252-259.
10. Ali M, Islam MZ, Sultana N. Prevalence of hypertension, diabetes, and dyslipidemia, and their associated factors: ATM study. Asian Trop Med. 2019;12(2):45-51.

11. Mukti BH. Sample size determination: Principles and applications for health research. Health Sciences International Journal. 2025;3(1):127-143. https://doi.org/10.71357/hsij.v3i1.63

12. Lwanga SK, Lemeshow S. Sample size determination in health studies: A practical manual. Geneva: World Health Organization; 1991.

13. Naing L, Winn T, Rusli BN. Practical issues in calculating the sample size for prevalence studies. Arch Orofac Sci. 2006 Jul;1:9-14.

14. American Diabetes Association. 2. Classification and diagnosis of diabetes: Standards of Medical Care in Diabetes—2025. Diabetes Care. 2025 Jan;48(Suppl 1):S6–17. doi:10.2337/dc25-S002.

15. Williams B, Mancia G, Spiering W, Agabiti Rosei E, Azizi M, Burnier M, et al. 2018 ESC/ESH Guidelines for the management of arterial hypertension. Eur Heart J. 2018;39(33):3021–3104.

16. Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults. Third Report of the National Cholesterol Education Program (NCEP) Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III) final report. Circulation. 2002 Dec 17;106(25):3143-421. https://doi.org/10.1161/circ.106.25.3143

17. Virani SS, Newby LK, Arnold SV, et al. 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA Guideline for the Management of Patients With Chronic Coronary Disease: A Report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines. *Circulation.* 2023;148(9):e9-e119. doi:10.1161/CIR.0000000000001168

18. Levin, K. Study design III: Cross-sectional studies. *Evid Based Dent* **7**, 24–25 (2006). https://doi.org/10.1038/sj.ebd.6400375

19. Sedgwick P. Convenience sampling. BMJ. 2013;347:f6304. https://doi.org/10.1136/bmj.f6304

20. Althubaiti A. Information bias in health research: definition, pitfalls, and adjustment methods. J Multidiscip Healthc. 2016;9:211–217. https://doi.org/10.2147/JMDH.S104807

21. Setia, Maninder Singh. Methodology Series Module 3: Cross-sectional Studies. Indian Journal of Dermatology 61(3):p 261-264, May–Jun 2016. | DOI: 10.4103/0019-5154.182410

22. Grimes DA, Schulz KF. Descriptive studies: what they can and cannot do. Lancet. 2002;359(9301):145–149. https://doi.org/10.1016/S0140-6736(02)07373-7

10. Declaration

I solemnly pledge that this research protocol shall be implemented in accordance with the relevant ordinance/circulars of relevant IRB.

I hereby declare that no part of the proposed research has been used in any thesis/dissertation in partial fulfilment of any degree/fellowship or any publication.

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Date **Dr. Sanjoy Kumar Ray**

Appendix-A

Title: "Prevalence of Dyslipidemia and Hypertension among Type 2 Diabetes Mellitus (T2DM) Patients in a Coastal Region of Bangladesh"

Data Collection sheet

Researcher's name: Dr. Sanjoy Kumar Ray **Date of data collection:**

Section A: Participant Information

1. **Patient ID:** ___________ **Contact mobile number (if any):** ________________
2. **Name:** __________________________ **Age (in years):** _________
3. **Gender:** ☐ Male ☐ Female ☐ Other ☐ Prefer not to say
4. **Height (in cm):** ____________ **Weight (in kg):** __________ **BMI (Calculated):** ____________
5. **Waist circumference (in cm):** __________ **Smoking status**: ☐ Yes ☐ No ☐ Former
6. **Physical activity level**: ☐ Sedentary ☐ Moderate ☐ Active
7. **Education level:** ☐ No formal education ☐ Primary school ☐ Secondary school ☐ College ☐ University level of education
8. **Monthly Family Income (in BDT):**
   ☐ <10,000 ☐ <20,000 ☐ <50,000 ☐ >50,000 ☐ Prefer not to say
9. **Occupation:**
10. **Residence:** ☐ Rural ☐ Urban

Section B: Diabetes mellitus status

11. **Duration of Diabetes (in years): _____________**
12. **What type of antidiabetic medications are you on?**

    ☐ *Oral antidiabetic agents* ☐ *Insulin* ☐ *Both* ☐ *None*
13. **Recent HbA1c (last 3 months): _______________** % *(if not available, collect blood sample for testing)*
14. **FBS/2HABF/RBS: ______________________** mmol/L
15. **Any feature of diabetic neuropathy (e.g., burning, numbness, tingling in limbs)?**

    ☐ Yes ☐ No

Section C: Blood Pressure Status

16. **Blood pressure at examination (mmHg):**
    - **Systolic: _____________ Diastolic: _____________**
17. **Do you have a history of hypertension (high blood pressure)?**

    ☐ Yes ☐ No

<u>If yes, fill the following question 18- 21</u>

18. **Duration of Hypertension in years: ______________**
19. **Are you currently on any antihypertensive medications?**

    ☐ Yes ☐ No
20. **Do you take your antihypertensive medication regularly?**

    ☐ Yes ☐ No
21. **Who did prescribe your anti-hypertensive medication?**

    ☐ Registered physician ☐ Nurse ☐ Health assistant (HA) ☐ Village practitioner/Quack
22. **When was your blood pressure last measured?** about ___________ months/years ago
23. **How frequently do you check your blood pressure?**

    ☐ Weekly ☐ Monthly ☐ 3 monthly ☐ 6 monthly ☐ Yearly ☐ Others

Section D: Dyslipidemia status

24. Are you currently taking any lipid-lowering medication (e.g., statins)?

☐ Yes ☐ No

25. **If yes: How long? _______________** months/years

26. **Lipid profile:** *(To be filled based on lab results)- **mg/dl:***

- **Total Cholesterol:** _________________ (High Total Cholesterol: >200 mg/dL)
- **LDL-C :** _________________ (High LDL-C: >130 mg/dL)
- **HDL-C :** _________________ (Low HDL-C: <40 mg/dL (men), <50 mg/dL (women))
- **Triglycerides :** ________________ (High Triglycerides: >150 mg/dL)

Section E: Comorbidity Overview

17. **Do you have any of the following problem?** (self-reported)

☐ Stroke

☐ Heart disease

☐ Kidney disease

☐ Thyroid disease**:** (☐ Hypothyroidism ☐ Hyperthyroidism)

☐ PCOS

☐ Fatty liver disease

18. Are you taking any other medication? (example: antiplatelet, uric acid lowering agent etc)

- List all the medications (if any):

Thank you for your co-operation and participation

Appendix-B

Written Informed Consent Form

Title of the study: Prevalence of Dyslipidemia and Hypertension among Type 2 Diabetes Mellitus (T2DM) Patients in a Coastal Region of Bangladesh:

1. Investigator's name: Dr. Sanjoy Kumar Ray
2. Institution/organization: Satkhira Medical College Hospital, Satkhira
3. Selection of the participant: Diabetes mellitus patients attended in Satkhira Medical College Hospital OPD.
4. Expectation from and involvement of the participant: History taking, performing clinical examination and relevant investigations including FBS, RBS, HbA1C, lipid profile.
5. Risks and benefits: The study will not cause any risk to the participants.
6. Privacy, anonymity and confidentiality: All of your personal information will be kept secret and will not be revealed any person or place other than the investigator & any person or place related with the study.
7. Right to withdraw: The patient has full right to withdraw himself/herself from the study at any time during study.

....

Signature or left thumb impression of the participant Signature of investigator

....

Signature or left thumb impression of a witness Date:

<u>অবগতিক্রমে সম্মতিপত্র</u>

Title: Prevalence of Dyslipidemia and Hypertension Among Type 2 Diabetes Mellitus (T2DM) Patients in A Coastal Region of Bangladesh

আমি সজ্ঞানে এবং স্বেচ্ছায় ডা. সঞ্জয় কুমার রায় পরিচালিত উপরোক্ত শিরোনামের গবেষণা কার্যক্রমে অংশগ্রহণ করিতেছি। ইহাতে আমাকে কোন আর্থিক সুবিধা প্রদান করা হইবেনা। তিনি এই বলিয়া আশ্বস্ত করিয়াছেন যে, এই গবেষণায় অংশগ্রহণে আমার শারীরিক অথবা মানসিক কোন ক্ষতি হইবেনা এবং যে কোন সময় আমি স্বেচ্ছায় ইহা হইতে অবমুক্ত হইতে পারিব এবং ইহাতে আমার চিকিৎসায় কোন প্রভাব থাকিবে না। আমার সমুদয় তথ্য অত্যন্ত গোপনীয় থাকিবে।

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রোগীর স্বাক্ষর	সাক্ষীর স্বাক্ষর	সাক্ষাতকারগ্রহণকারীর স্বাক্ষর
তারিখঃ	তারিখঃ	তারিখঃ